

A PROJECT REPORT

Submitted by

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Sem – 6th

Degree ENGINEERING

in

Information and Communication Technology Engineering



Faculty of Technology

Marwadi University, Rajkot

Information and Communication Technology

2021-22

CERTIFICATE

This is to certify that the project entitled **Chat Application** has been carried out by **Dhanrajsinh Parmar (92110133001)** under my guidance in partial fulfillment of the B.tech Engineering in Information and Communication Technology (6th Semester) of Marwadi University, Rajkot during the academic year 2023-24.

Date : 08-05-2023

Internal Guide

Head of the Department

Prof. Shamsagazarzoo Alam

Prof. C. D. Parmar

Assistant Professor

ICT Engineering

Information and Communication Technology

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Introduction

1.1 Document Purpose

The document purpose is to provide a detailed description of the single and group chat application, outlining its objectives, requirements, and constraints to ensure the development team and stakeholders have a clear understanding of the project's scope and deliverables.

1.2 Product Scope

The single and group chat application aims to provide a platform for users to communicate in real-time using text messages. The application will allow users to create their profiles, search for other users, and add them to their contact list. Users can initiate one-on-one or group chats with their contacts and engage in conversations in real-time. The application will have features such as user authentication, message history, notifications, and the ability to create and join chat rooms. The application will be developed using React.js and Node.js with socket communication to enable real-time messaging between users. The application's primary focus is to provide a user-friendly, secure, and reliable platform for single and group chatting, with a user interface that is simple and easy to navigate.

1.3 Acronyms/Abbreviations

Abbreviation:

- React.js, Node.js, API, SQL, HTML, CSS, JS, JSON
- MongoDB

1.4 Document Conventions

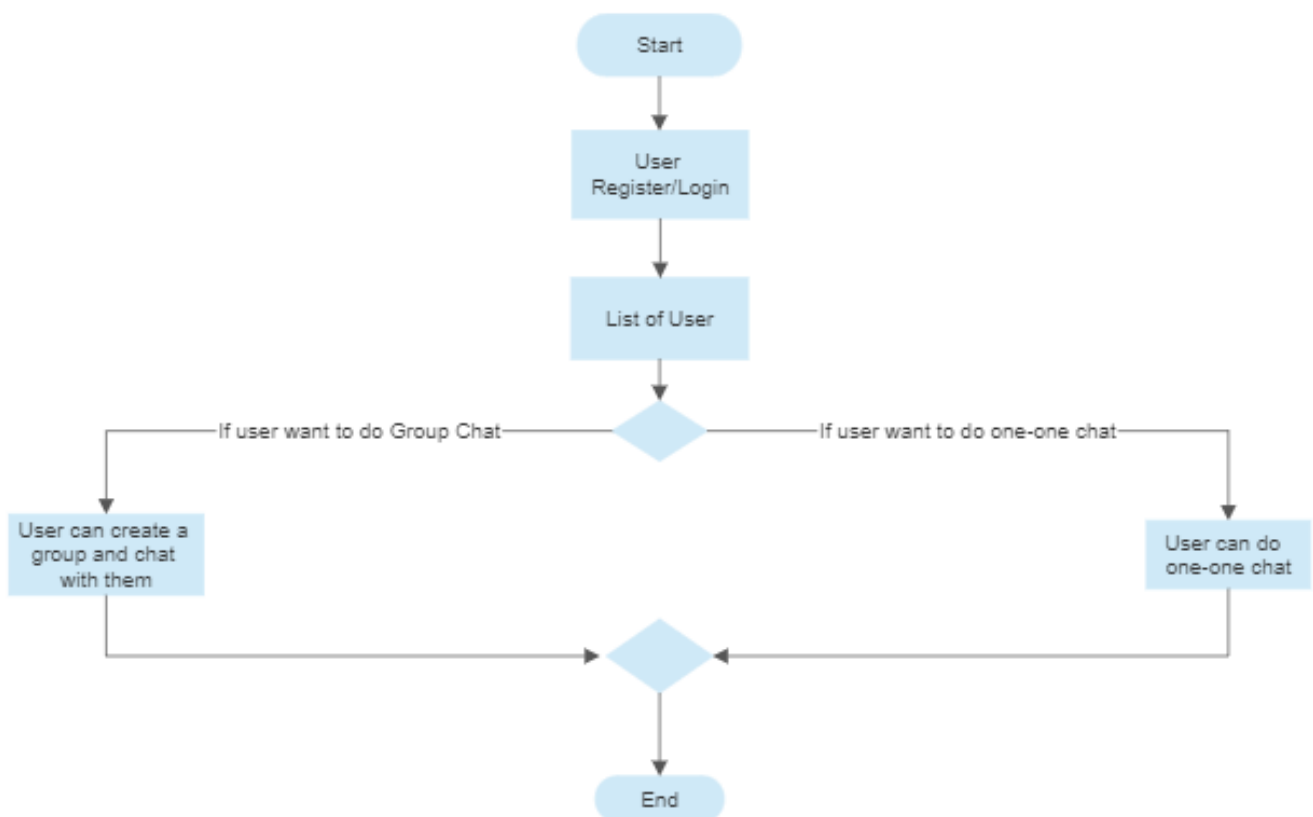
Use Calibri font size 11, or 12 throughout the document for text. Document text should be single-spaced and maintain the 1" margins found in this template.

2. System Design Considerations

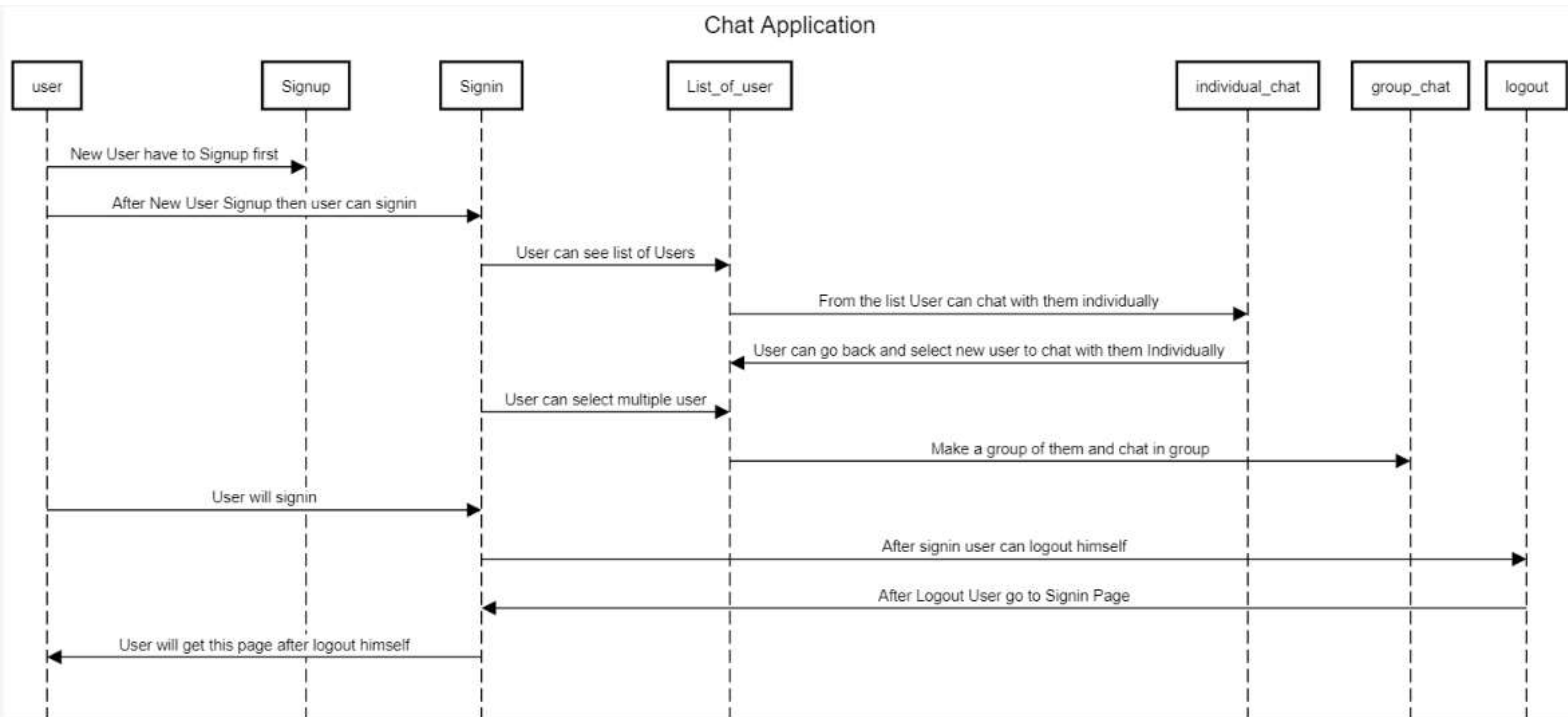
2.1 Exception Handling

The chat application should have robust exception handling to ensure smooth functioning and error-free execution. The application should be able to handle errors gracefully, such as server-side errors, network errors, and input validation errors, among others. The application should have a well-defined error handling strategy, such as logging errors, displaying user-friendly error messages, and informing the user about the issue and potential solutions. The application should also have a mechanism to handle exceptions related to the use of third-party APIs and services. The development team should perform extensive testing to identify and address potential exceptions and errors, ensuring that the application meets the highest quality standards.

2.2 Flowchart



2.3 Sequence Diagram



3. Overall Description

3.1. Product Perspective

The single and group chat application will be a standalone web-based application, accessible through a web browser or a mobile device. The application will communicate with a back-end server using RESTful APIs, enabling real-time messaging between users. The application will be developed using React.js and Node.js, which are modern and popular web development frameworks that allow for efficient development and easy maintenance. The application will have a user-friendly and responsive UI, with a simple and intuitive design that is easy to navigate. The application will also have a back-end server that will handle requests, manage user accounts, and store message history data securely. The application will be scalable and robust, capable of handling a large number of users and messages simultaneously. Overall, the application will provide an efficient and reliable platform for single and group chatting, with a focus on user experience and security.

3.2. Product Functionality

The single and group chat application will provide the following core functionalities:

- **User Registration and Login:** Users will be able to create their profiles, with a unique username and password. They can then use their login credentials to access the application.
- **One-on-One Chat:** Users will be able to initiate one-on-one conversations with their contacts, where they can exchange text messages in real-time.
- **Group Chat:** Users will be able to create and join chat rooms with multiple users and participate in group conversations.
- **Contact List:** Users will be able to search for other users and add them to their contact list for easy access.
- **Message History:** Users will be able to view their chat history, including one-on-one and group conversations.
- **Security:** The application will have secure authentication mechanisms to ensure that only authorized users can access the application and communicate with each other.
- **Error Handling:** The application will have robust error handling mechanisms to ensure smooth functioning and error-free execution.
- **Search:** Users will be able to search for messages, users, and chat rooms to quickly find the information they need.

3.3. Operating Environment

The operating environment for website using React.js, Node.js and MongoDB is as listed below.

3.3.1. **Operating System:** The application will be developed and tested on Windows operating systems, including Windows 7, 8, and 10.

3.3.2. **Integrated Development Environment (IDE):** The development team will use Visual Studio as the primary IDE for development and testing.

- 3.3.3. Programming Languages: The application will be developed using React.js, Node.js, JavaScript (JS), TypeScript (TSX), HTML, CSS, and MongoDB.
- 3.3.4. Frameworks and Libraries: The application will be built using various frameworks and libraries, including Express.js, Socket.IO, Mongoose, and Bootstrap.
- 3.3.5. Database: The application will use MongoDB as the primary database to store user information and message history data.

3.4. Assumptions and Dependencies

The development team assumes that the end-users will have basic knowledge of using a web browser and internet connectivity to access the application. The team also assumes that users will have valid email addresses to register and access the application. Additionally, the team assumes that the application will be accessed by users with various devices, including desktops, laptops, tablets, and smartphones.

The successful development and deployment of the single and group chat application depend on various factors, including:

1. Technology Dependencies: The application relies on various technologies, such as React.js, Node.js, MongoDB, and Socket.IO, which must be installed and configured correctly.
2. Third-Party APIs: The application uses third-party APIs, such as Google Maps and Facebook, which must be integrated correctly.
3. Internet Connectivity: The application requires a stable and reliable internet connection for real-time messaging.
4. Browser Compatibility: The application must be compatible with various web browsers, including Chrome, Firefox, Safari, and Edge.
5. User Input: The application relies on user input, which must be validated to prevent security vulnerabilities.
6. Security: The application must be secure and protect user data from unauthorized access and malicious attacks.
7. Testing: The application must be thoroughly tested to ensure that it meets the highest quality standards and performs as expected.

3.5. Functional Requirements

The single and group chat application must meet the following functional requirements:

1. User Registration: The application must allow users to create a new account by providing their name, email address, and password.
2. User Login: The application must allow registered users to log in to their accounts using their email address and password.
3. One-on-One Chat: The application must allow users to initiate one-on-one conversations with their contacts and exchange text messages in real-time.
4. Group Chat: The application must allow users to create and join chat rooms with multiple users and participate in group conversations.
5. Contact List: The application must allow users to search for other users and add them to their contact list for easy access.

6. Message History: The application must allow users to view their chat history, including one-on-one and group conversations.
7. Security: The application must use secure authentication mechanisms to ensure that only authorized users can access the application and communicate with each other.
8. Error Handling: The application must have robust error handling mechanisms to ensure smooth functioning and error-free execution.
9. Search: The application must allow users to search for messages, users, and chat rooms to quickly find the information they need.

4.Demonstrate Project

Sign Up Page: -

Chat Application

Login Sign Up

Name *

Enter Your Name

Email *

Enter Your Email

Password *

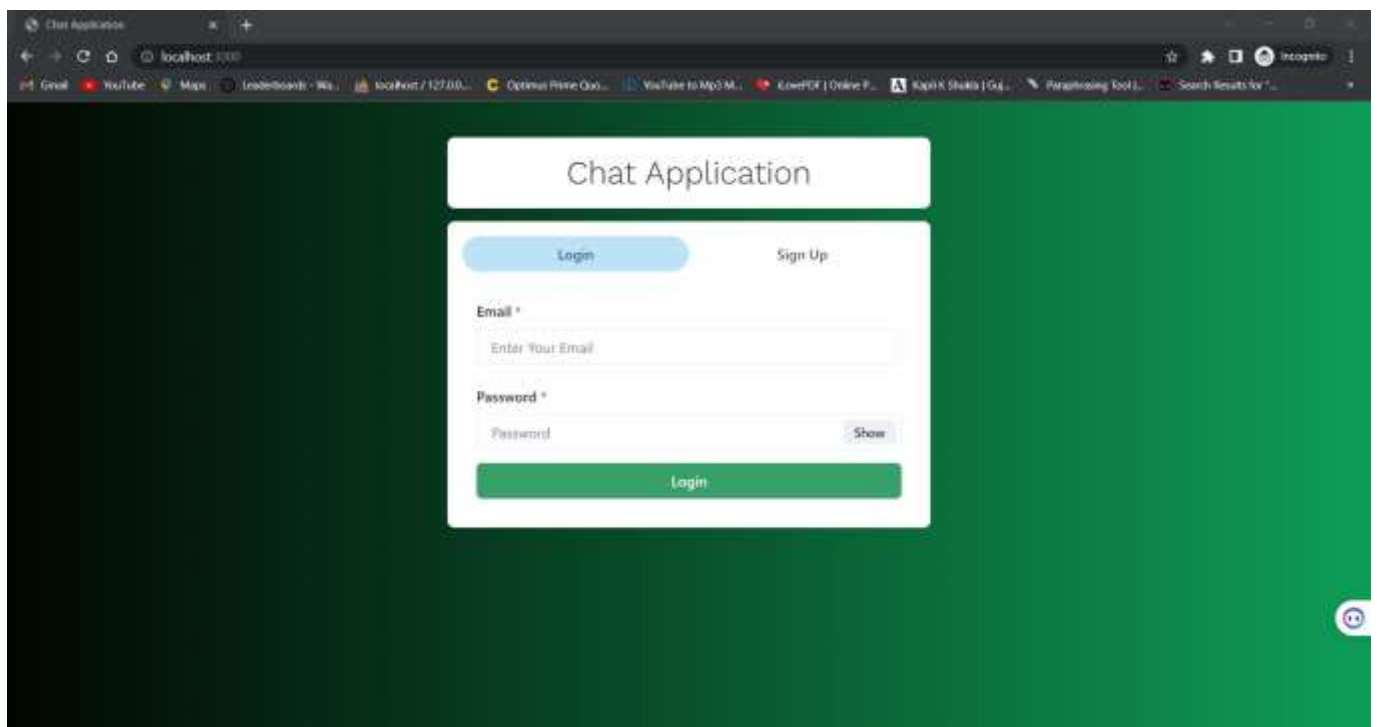
Password Show

Confirm Password *

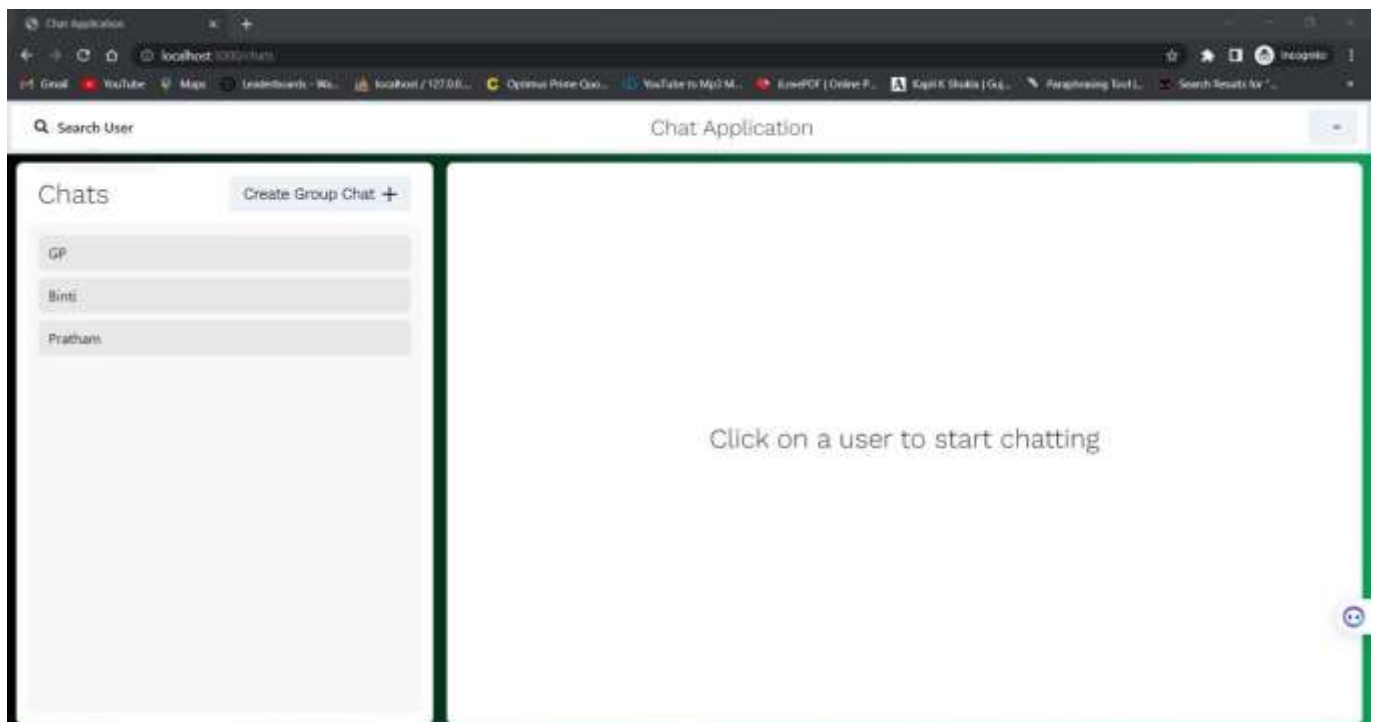
Confirm Password Show

Sign Up

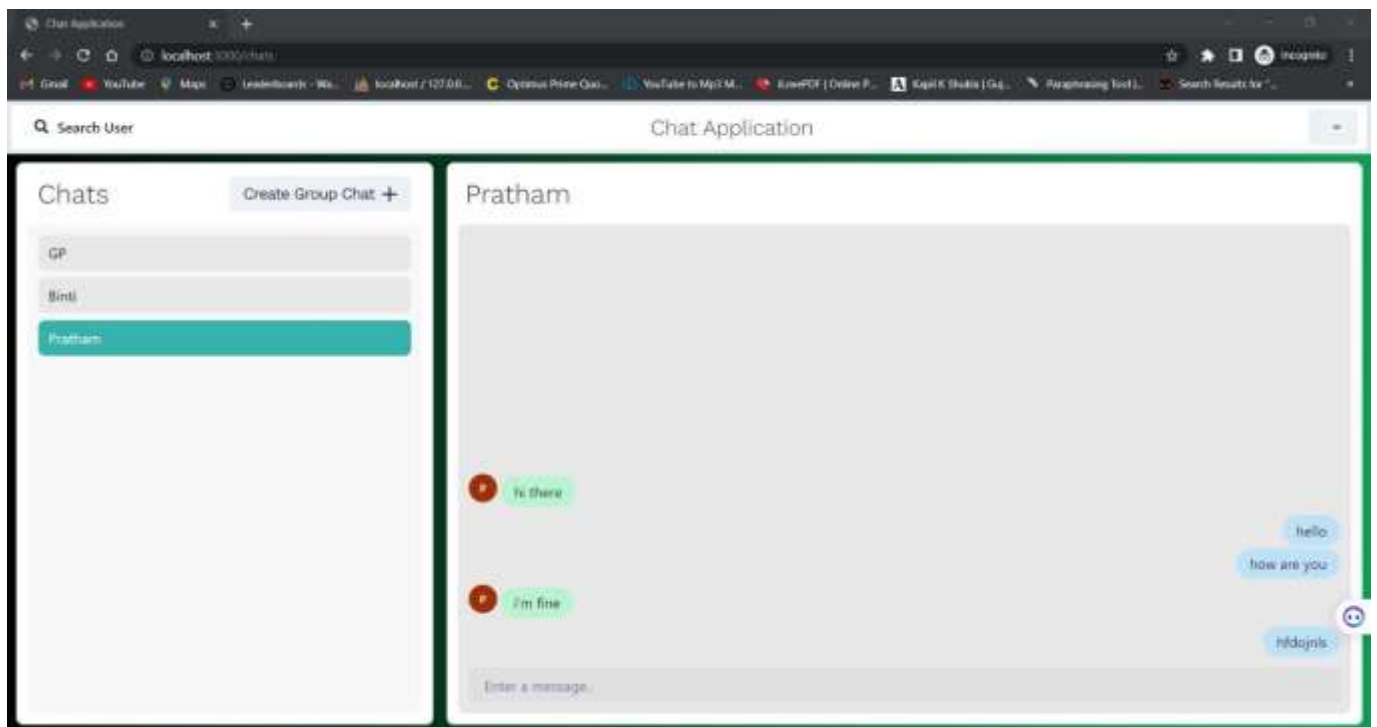
Sign in Page: -



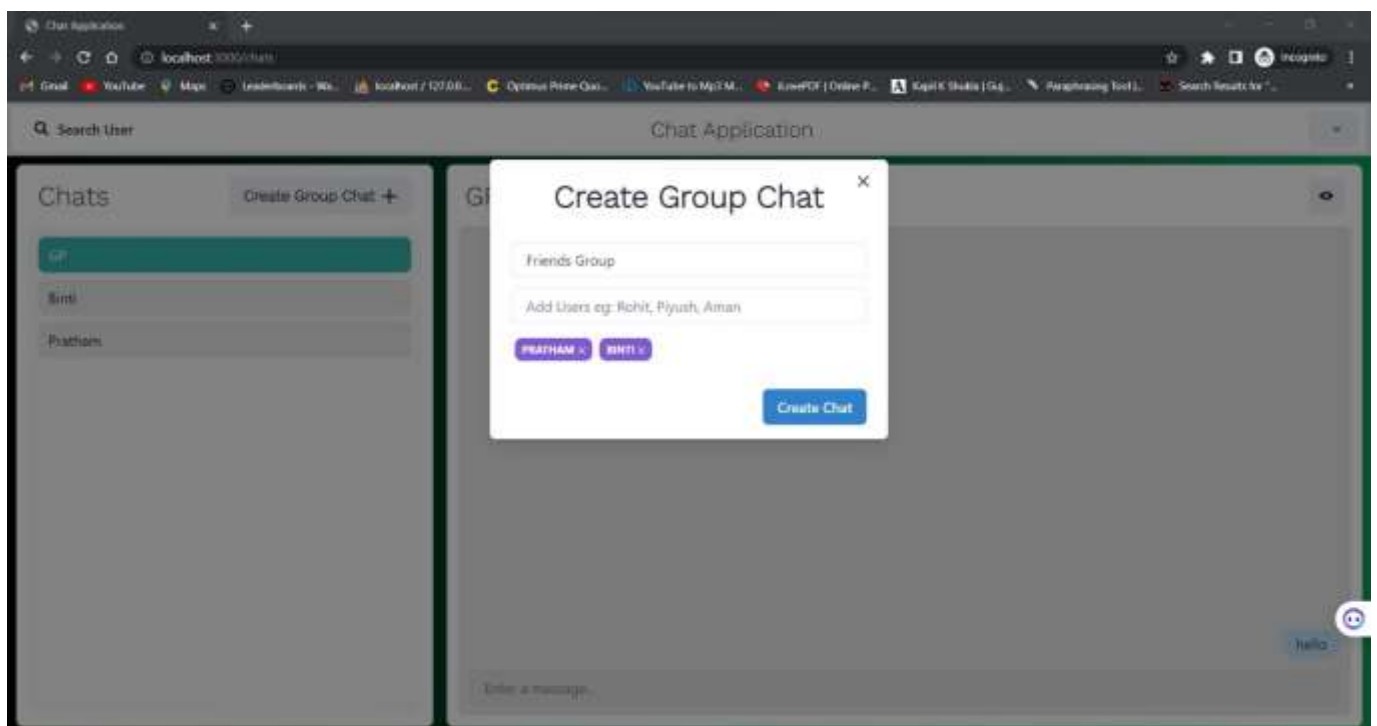
Select User For Chat: -



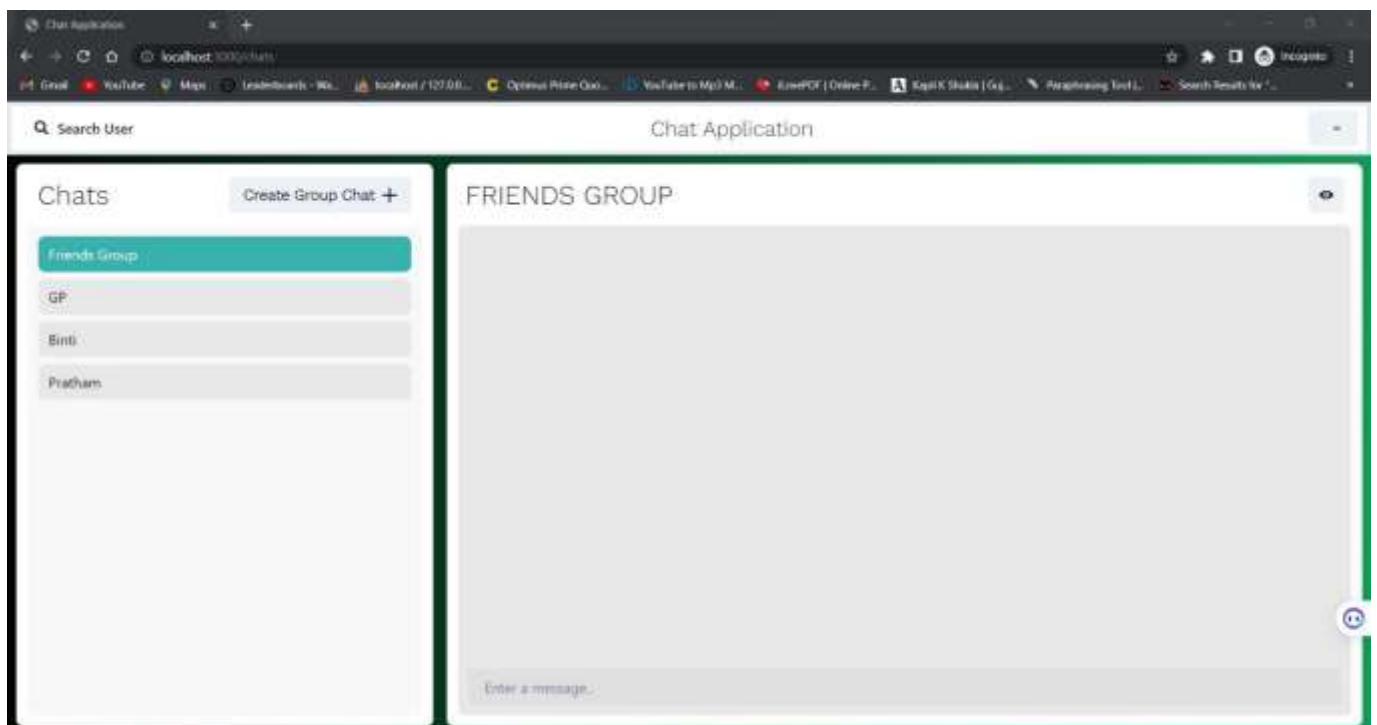
Individual Chat: -



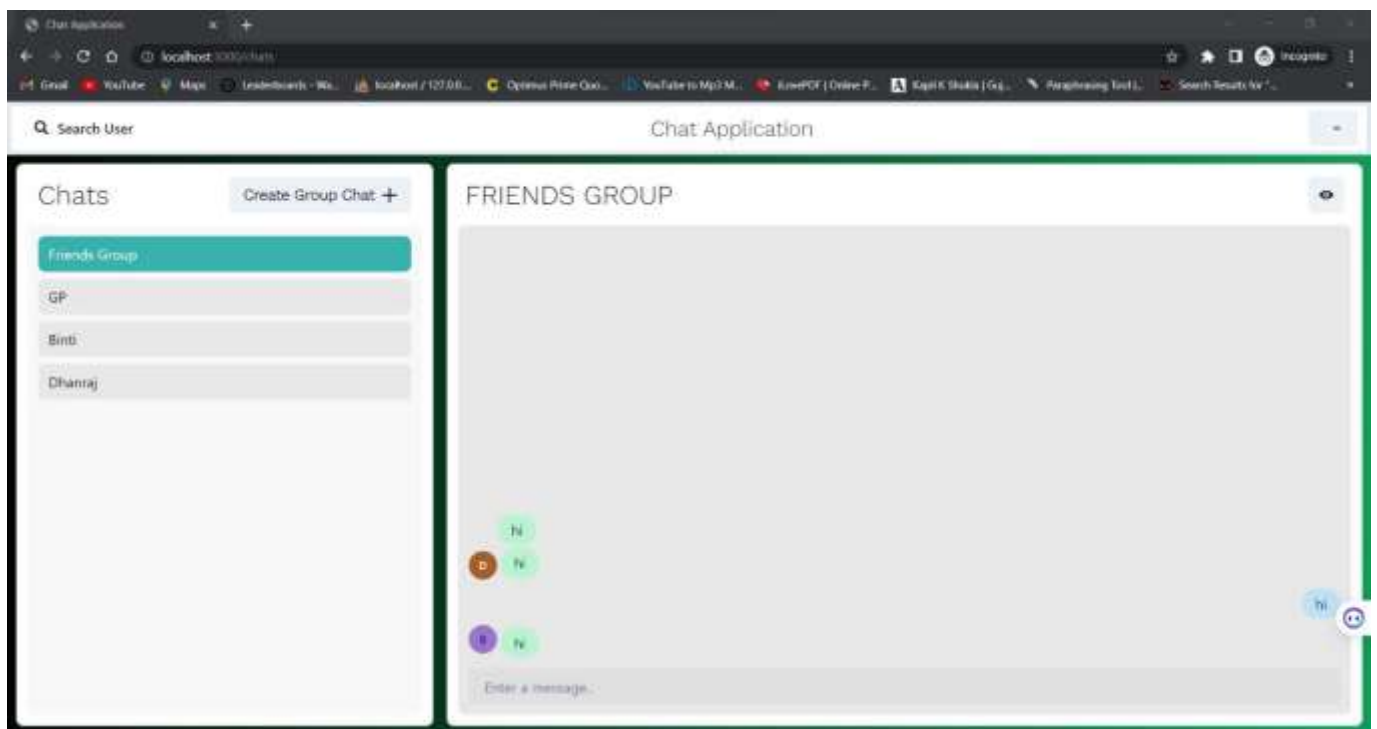
Create Group for group chat: -



Group: -



Group Chats: -



4.2. Apache Configuration

MongoDB is a popular NoSQL document-oriented database that stores data in flexible, JSON-like documents and allows for easy scalability and high availability.

